A Robust WiFi Localization Algorithm Using Data Augmentation and Stacked Denoising Autoencoder

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***Abstract*—WiFi-based fingerprint localization technology has become one of the most practical methods for localizing mobile users due to its non-intrusive nature, low cost and no additional equipment required. However, the fluctuation of WiFi signal seriously affects the accuracy of WiFi fingerprint localization. To address this problem, this paper proposes a solution using data augmentation combined with stacked denoising autoencoder (SDAE). Data augmentation can facilitate the neural network to learn the mapping relationship between the fluctuating WiFi signals and coordinates. And the SDAE can obtain a robust and time-independent feature from the dynamic WiFi signal. A convolutional neural network is also used to build a floor classification model to determine the height, and a multilayer perceptron (MLP) is used to build a regression model to determine the relative coordinates. Experimental results on public datasets show that the method improves system robustness and localization accuracy.**

Keywords—indoor positioning, data augmentation, stacked denoising autoencoder.

I. Introduction

The demand for indoor positioning, such as location-based wireless advertising push, information retrieval and pedestrian navigation, is growing rapidly [1]. Currently, due to the wide deployment and availability of WiFi infrastructure, WiFi fingerprint-based localization is low-cost and does not require the deployment of additional hardware [2]. It has become one of the most dominant indoor positioning techniques. The signal vector composed of multiple WiFi access points (APs) measured at a certain location can be used as a fingerprint to tag that location. When performing localization, the measured WiFi signals are used to estimate the location of the user or device by finding the closest match in the database using a matching algorithm [3].

Driven by the development of artificial intelligence, deep learning techniques have been widely used to process wireless signals. Unlike traditional methods including K-NearestNeighbor (KNN) [4], support vector machine (SVM) [5] and probabilistic inference [6], which are computationally intensive and time-consuming and require complex filtering and parameter tuning. Deep learning can efficiently process large amounts of fingerprint data, extract reliable features, and construct internal representations from dynamic signals without additional human intervention [7].

Currently, WiFi signals used for fingerprint localization include Receiving signal strength (RSS) and Channel Status Information (CSI). CSI is more stable and informative compared to RSS signals. Neural networks using CSI can achieve very accurate localization performance [8]. In addition, each antenna can receive multiple subcarriers, so a single AP can be used for localization [9]. However, CSI-enabled devices are not yet popular, and CSI-based localization methods are still in the laboratory stage. The current mainstream still uses RSS for fingerprint localization.

Ideally, the received signal strength is proportional to the distance, but the complexity of indoor scenes and interference from other wireless devices cause signal volatility, which negatively affects localization accuracy. The goal of deep learning is to extract reliable features from the fluctuating signals and to find an accurate mapping function of signal fluctuation intervals to locations.

In [10], a floor classification model is proposed. Stacked autoencoder (SAE) is used for dimensionality reduction and feature extraction of sparse RSS fingerprint data. In [11], a DNN architecture for multi-label classification is proposed to solve the scalability problem for multi-class classification tasks, and the neural network structure consists of SAE and feedforward neural networks. In [12], RSS-based fingerprint data is converted into images and then a CNN model is used to improve the robustness of the localization system to changes in the input signal. In [13], a deep learning model for multilevel localization is proposed. And for the first time, SAE is combined with a one-dimensional CNN model for floor classification and coordinate regression. In [14], SDAE is introduced to further improve the robustness of the system and mitigate the negative impact of signal volatility. In [15], to minimize the impact of signal fluctuations, the signal is sorted according to its strength and a corresponding AP sequence is generated, which is then processed as a sequence using a recurrent neural network.

The WiFi signal has volatility, especially when the time interval is longer, the fluctuation is more intense. To maintain good positioning performance, it is necessary to periodically collect new samples and train the positioning model, but this is time-consuming and labor-intensive. This paper proposes a solution using data augmentation and SDAE to mitigate the negative effects of signal fluctuations. The performance of this method is proved to be effective by comparing with other methods.

The rest of this paper is organized as follows. In Section II, the WiFi fingerprint-based indoor localization system is introduced, including the method of data augmentation and SDAE, etc. In the Section III, the effectiveness of the method is verified by experiments on a public dataset. Finally, the conclusion section of the whole paper is given.

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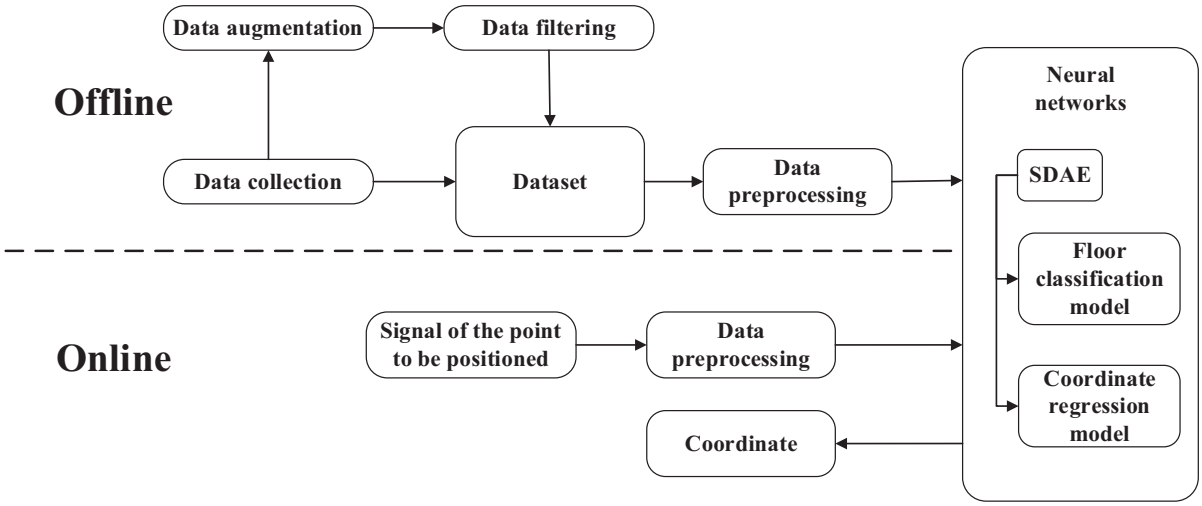

Fig. 1. Architecture of indoor positioning system based on WiFi fingerprint.

II. SYSTEM DESIGN

A. System Overview

The architecture of WiFi fingerprint-based indoor localization system is shown in Fig. 1, in which the localization process includes an offline training phase and an online localization phase. In the offline phase, fingerprint data collection and data augmentation are performed to build a fingerprint database, and then the training of the neural network model is completed. The trained neural network model is used in the online phase to match the received fingerprint measurements to estimate the user's location.

B. Data Collection

The fingerprint data is composed of the RSS measurements of AP measured at indoor reference point. Considering the common range of raw RSS observation from -110 dBm to 0 dBm, the missing RSS value of WiFi signal is set to -110 dBm in this paper. Fingerprint data needs to be tagged according to the actual location of the reference point, such as building ID, floor ID, latitude and longitude, in preparation for the training of the neural network.

$$\{\mathrm{rss}_0, \mathrm{rss}_1, \cdots, \mathrm{rss}_n, floor, x, y, Timestamp\} \quad (1)$$

The fingerprint data format is shown in (1), where rss_i is the RSS measurement value of the AP at the reference point. The latitude and longitude are represented by relative coordinates (x, y).

C. Data augmentation

WiFi signals are volatile and the fluctuations are more intense at longer time intervals. This requires periodically collecting new samples to expand the fingerprint database to ensure the accuracy of the positioning system. In this paper, we simulate WiFi signals according to the nature of RSS distribution and realistic data enhancement to avoid time-consuming duplication of labor. The expanded dataset can fully express the mapping relationship between the RSS fluctuation interval corresponding to each AP of the fingerprint data and the location, enabling the neural network to learn time-independent robust features to improve the robustness of the system.

Usually fingerprint data are formed by collecting RSS measurements at the same moment for each AP at the reference point. However, the RSS values fluctuate irregularly with time and the APs are independent of each other. Therefore, in this paper, the data is processed in the unit of AP to achieve data augmentation. The specific steps of data augmentation are described below.

First, the fingerprint data is classified according to the reference point ID. After the classification is completed, the RSS signal is then collected according to the AP to form an set. A reference point will form multiple sets. Then, in each set, a value is randomly selected to construct a new fingerprint data. This ensures that the signal distribution of reference points from a single AP remains unchanged, while expanding the range of combinations between APs.

The RSS distribution shows a left-skewed normal distribution. The fingerprint data can be further extended by fitting a normal distribution function based on RSS_APi to simulate signal fluctuations. First, the proportion of missing values in the set RSS_APi is calculated as r, and calculate the mean μ and standard deviation σ after removing missing values. A normal distribution N (μ, σ^2) is fitted according to the mean and standard deviation as the RSS distribution model for the reference point. Then, when generating data, generate a random number *rand* between 0 and 1, if $rand < r$, set the RSS value to -110, otherwise generate a random number as the RSS value according to the normal distribution N (μ, σ^2). According to the above method, the RSS value of each AP in the reference point is generated to form new fingerprint data.

Since the sampling environment is simpler compared to the real scene, the mean value of RSS in the real scene is generally smaller than that in the sampling environment. To simulate the real scene, the mathematical expectation of the fitted normal distribution can be reduced by δ, and the standard deviation is kept constant to generate some additional data as a supplement.

When generating data, the above 3 methods generate twice the original data in the ratio of 2:1:1. The distribution is

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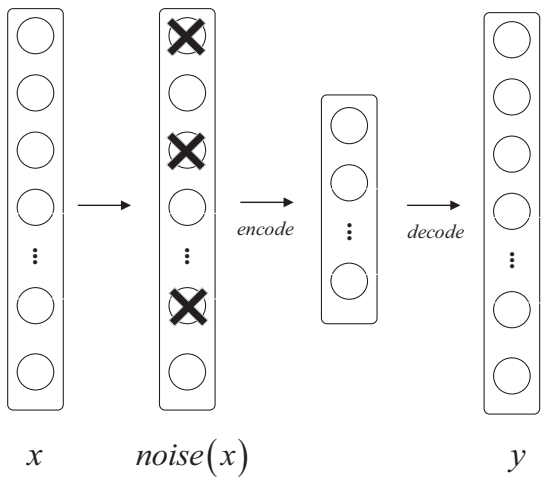

Fig. 2. DAE neural network structure with one hidden layer.

extended while keeping the original RSS distribution unchanged.

D. Data Filtering

In this paper, the discriminator of GAN is trained as a filter, and both the generator and the discriminator of GAN model are built by a simple linear neural network with 3 layers. The generator generates realistic data and the discriminator is trained with high quality negative samples in the adversarial process. The problem of missing negative samples for a class of classifiers is circumvented using GAN [16].

E. Dataset

This paper uses the publicly available indoor positioning dataset UJIIndoorLoc [17] for neural network modeling. The UJIIndoorLoc dataset contains 21049 fingerprint samples covering 3 buildings with 4 to 5 floors, 933 reference points, and a coverage area of 108703 m2. Each sample contains 520 RSS measurements ranging from -110 dBm to 0 dBm. Also includes reference point location information such as building and floor numbers, latitude and longitude, time stamps, and user and device tags. The average number of samples per reference point is about 23, and each RP detects an average of 18 APs. There are 1111 samples collected after the fourth month after the data set was built, and the collection points are not repeated with the reference points and are used as the test set.

F. Data Pre-processing

Direct use of raw data leads to neural network gradient explosion and difficult convergence of the loss function. Data pre-processing is needed to provide reasonable input for the feature extraction module. We normalize the raw RSS data to the range between 0 and 1 using (2). By employing this normalization operation, the distribution of normalized WiFi data is unbiased and has low variance.

$$rss_i = \frac{(rss_i - \min_rss)}{(\max_rss - \min_rss)} \tag{2}$$

G. SDAE-based Feature Extraction

To accurately fit the mapping relationship between RSS signal fluctuation range and location, this paper improves SDAE according to the nature of RSS, and extracts time-independent robust features from dynamic fingerprint data. The dynamic fingerprint data is mapped to a low-dimensional stable feature vector, thereby alleviating the negative impact of signal fluctuations on positioning accuracy.

Denoising autoencoder (DAE) is a type of autoencoder (AE). AE is a neural network that uses a backpropagation algorithm to make the output value equal to the input value. the neural network structure of AE consists of two symmetric parts, i.e., the encoder and the decoder. the DAE adds noise to the original input on top of the AE, and then reconstructs the original input data through the autoencoder, forcing the encoder to learn and extract important robust features from the input data. The SDAE adds noise in a similar way to the Dropout layer by randomly setting a certain percentage of the input to zero and corrupting the data. The neural network structure of DAE with one hidden layer is shown in Fig. 2, which includes three parts, noise adding part, encoder and decoder.

SDAE is a training method of autoencoder, and SDAE uses layer-by-layer training to minimize the reconstruction error. In this paper, the proposed SDAE neural network structure consists of three stacked DAEs. Its training process is shown in Figure 3, and the specific process is described as follows: 1) Given an initial input, an autoencoder with one hidden layer is trained in an unsupervised manner until the reduced reconstruction error reaches a threshold value. 2) Train the autoencoder by taking the output of the previous autoencoder hidden layer as the input of the current autoencoder. 3) Repeat step 2 until all autoencoders have been trained. 4) Use the output of the hidden layer of the last autoencoder as the features extracted from the original data.

The traditional SDAE directly uses the Dropout layer as the noise addition method, forcing the encoder to learn and extract robust features from the damaged data. However, the fingerprint data is sparse. Taking the UJIIndoorLoc dataset as an example, the length of the fingerprint data is 520, the average effective signal is 18, and there are only 18 non-zero values on average after data preprocessing. If the Dropout layer is used to randomly set the data to 0, the training effect will be unstable. In addition, setting 0 means directly changing a valid signal to an invalid signal, for example, setting −10 dBm to −110 dBm. Such a large-scale fluctuation does not conform to the actual situation, and some information may be

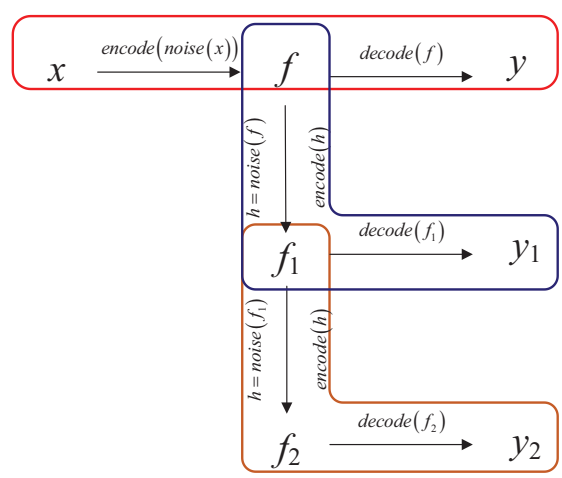

Fig. 3. SDAE training process.

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lost, affecting the positioning accuracy of the regression model.

This paper adds appropriate noise to all effective signals of the input fingerprint data to simulate signal fluctuations in real scenes. Use an autoencoder to reconstruct the original input from the noisy fingerprint data. The encoder is forced to learn to map dynamic fingerprint data into low-dimensional stable feature vectors, thereby alleviating the impact of signal fluctuations on positioning accuracy.

The RSS fluctuation range is generally within 20 dBm, and the value is randomly selected between [-10, 10], when adding noise to the effective signal. After normalization, the value is randomly selected between [-0.91, 0.91] to add noise. In the first DAE, a value between -0.91 and 0.91 is randomly added as noise to all valid signals to simulate signal fluctuations. The second DAE and the third DAE normally use the Dropout layer as a way to add noise, with parameters of 0.5 and 0.6, respectively. The activation functions of the three DAEs are all ReLu, and the number of neurons in the output layer of the encoder is 256, 128 and 64, respectively. After the training is completed, 64 time-independent robust features can be extracted using SDAE.

A suitable noise is added to the original WiFi data so that it can approximate the distribution of new WiFi data with a large time interval from the original WiFi data. SDAE is then used to extract stable features from the dynamic fingerprint data, which ultimately improves the localization accuracy and robustness of the system.

H. Floor Classification Model

In this paper, the combination of building ID and floor ID is used as the label of the floor classification model. The 64 features extracted by SDAE are used as the input of the classification model. The neural network structure of the floor classification model consists of three one-dimensional convolutional layers and three fully connected layers. The number of convolution kernels of the three one-dimensional convolutional layers is 32, 64, and 32; the convolution kernel sizes are 15, 9, and 5; the step size is 2, 1, and 1; the padding is 7, 4, and 2. The number of neurons in the three fully connected layers are 520, 128 and 32 respectively. A Dropout layer with a parameter of 0.5 is added between each layer to prevent overfitting. The entire network uses ReLu as the activation function and the cross-entropy function as the loss function.

I. Coordinate Regression Model

There are two types of methods to determine indoor coordinates using neural networks: one is to train a coordinate regression model using neural networks; the other is to train a reference point classification model using neural networks, and then combine the predicted values of the networks and perform a weighted KNN operation. Considering that the localization accuracy of the second method is very dependent on the reference point density, the regression model is used in this paper for the scalability of the model and the smoothness of the localization results.

The 64 features extracted by SDAE are used as the input of the coordinate regression model. Including 8 layers of neural network, the number of neurons are 128, 256, 256, 128, 64, 32, 16 and 8 respectively. Use ReLu as the activation function. A regularization layer is used to speed up the loss function convergence process while alleviating the overfitting problem. Since the range of relative coordinates is between 0 and 400 m, in order to speed up the convergence, the coordinates are normalized between 0 and 1. Correspondingly, set the output layer activation function to Sigmoid to map the output between 0 and 1. Use Mean squared error (MSE) as the loss function.

III. Experiments and Evaluation

In this section, this paper validates the superiority of the proposed data enhancement method and the improved SDAE by comparing previous work. To facilitate the comparison of the algorithms, the publicly available datasets UJIIndoorLoc and UTSIndoorLoc are used in this paper. in addition, to verify the robustness of the algorithms, a new dataset is partitioned from the UJIIndoorLoc dataset according to the acquisition time and is denoted as UJI_Time in this paper.

A. Experiments on UJIIndoorLoc Dataset

The average number of samples for reference points in the UJIIndoorLoc dataset is 23. Considering that the mean and standard deviation of a small number of samples may have large errors, it is difficult to accurately describe the distribution of RSS. When generating data for reference points with sampling number less than 25, the fingerprint data is segmented with AP as the unit, and then restructured without fitting the distribution model. For reference points with 25 or more samples, both methods can be used. The sampling environment is simpler than the real scene, and the mathematical expectation of the fitted normal distribution is higher. The mathematical expectation can be reduced by δ, and the standard deviation is kept constant to generate some additional data as a supplement. The impact of the value of δ on the positioning accuracy is shown in Figure 4. The floor hit

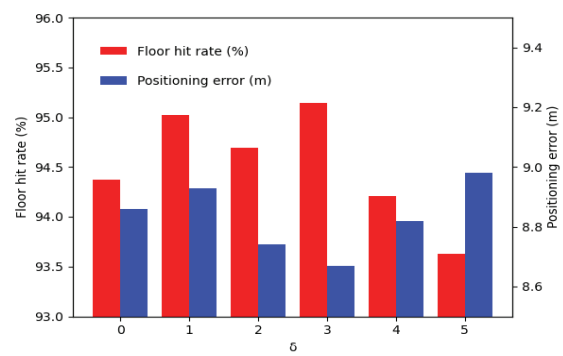

Fig. 4. The impact of δ on the positioning accuracy.

TABLE I. Neural Network Hyperparameters

Parameter	SDAE	Classification Model	Regression Model
Batch size	32	64	64
Activation	ReLu	ReLu	ReLu
Optimizer	Adam	Adam	RMSprop
Learning rate	0.01-0.00001	0.01-0.00001	0.01-0.00001
Epochs	200	150	200
Loss function	MSE	CrossEntropy	MSE

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TABLE II. PERFORMANCE COMPARISON OF METHODS ON UJIINDOORLOC DATASET

	Floor hit rate (%)	Positioning error (m)
Initial	93.78	9.42
Data augmentation	95.14	8.67
SDAE	94.26	9.73
Improved SDAE	95.05	9.05
Data augmentation and improved SDAE	96.04	8.37

rate shows an increasing and then decreasing trend, while the positioning error shows a decreasing and then increasing trend. The optimal effect is achieved when δ is set to 3, and the positioning accuracy drops significantly when δ is greater than 3.

The hyperparameter settings of the neural network are shown in Table 1. All three neural networks are trained 150 times. In order to ensure that the loss function fully converges, a learning rate decay method is adopted. The initial learning rate is 0.001, and the learning rate decays by 90% every 50 trainings.

The effect of methods on the UJIIndoorLoc dataset is shown in Table 2. Figure 5 shows the cumulative distribution function (CDF) of positioning errors of each method in Table 2. Experiments prove that the data augmentation method and the improved SDAE are effective. Without using data augmentation and SDAE, the floor classification model based on the convolutional neural network and the coordinate regression model based on the multi-layer perceptron can achieve a floor hit rate of 93.78% and an average localization error of 9.42 through parameter tuning. With data augmentation, both aspects are significantly improved. Using SDAE directly improves the floor hit rate to a certain extent, but the localization error of the regression model increases. This is because although SDAE can extract robust features, it may also lose some useful information, which helps to improve the accuracy of the classification model but has a negative impact on the regression model. This problem is solved by adding reasonable noise to the improved SDAE. Combining with data augmentation, the floor hit rate is ultimately increased by about 2.26%, the average localization error is reduced by 1.04m, and the accuracy is improved by 8.97%.

TABLE III. COMPARISON RESULTS WITH OTHER LITERATURE

Paper	Floor hit rate (%)	Positioning error (m)
[11]	92.89	9.29
[12]	95.41	-
[13]	96.03	11.78
[17]	89.92	7.9
[18]	93.8	7.4
[19]	93	8.63
This paper	96.04	8.37

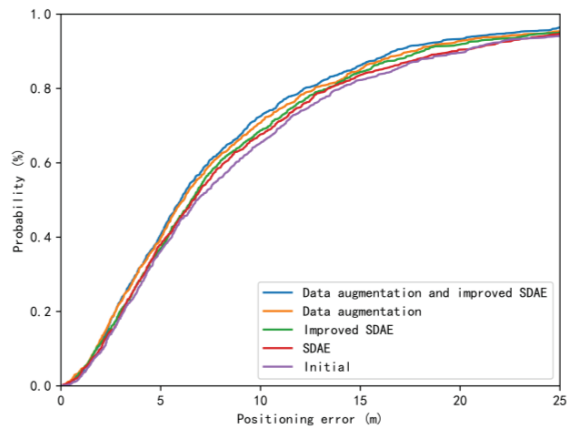

Fig. 5. The CDF of positioning errors.

Table 3 shows the results comparison between the best solution in this paper and methods in other literatures on the public dataset UJIIndoorLoc. The performance achieved in this paper is superior or comparable to other methods in this field. Although the positioning error has a certain gap compared with [17] and [18], the method proposed in this paper achieves the best in terms of floor hit rate. In addition, [17] and [18] uses the complete data set provided by the 2014 IPIN conference, including training set, verification set and test set. Because the verification set is not open, other papers use the data set, which only includes training set and verification set.

B. *Experiments on UJI_Time Dataset*

The UJI_Time dataset is split from the UJIIndoorLoc dataset to verify the robustness of the algorithm. The WiFi signal fluctuates with time, so this paper sorts UJIIndoorLoc according to the timestamp, and divides it into training data set, verification data set and test data set. The training data set was collected on May 30, 2013, and the test data set was collected on June 10, 2013.

Table 4 is a comparison of the method described in this paper and the method in [14] on the UJI_Time dataset. The fingerprint collection interval between the test set and the

TABLE IV. PERFORMANCE COMPARISON OF METHODS ON UJITIME DATASET

Paper	Positioning error (m)
[14]	5.64
This paper	4.13

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TABLE V. COMPARISON OF METHODS ON UTSINDOORLOC DATASET

Paper	Floor hit rate (%)	Positioning error (m)
[13]	94.57	7.60
[15]	94.3	6.54
This paper	95.36	7.25

training set on the UJI_Time dataset is 10 days. Compared with [14], the positioning error of the method proposed in this paper is reduced by about 1.5m, which verifies the robustness of the algorithm in this paper.

C. Experiments on UTSIndoorLoc Dataset

The UTSIndoorLoc dataset was collected in the FEIT Building at University of Technology Sydney (UTS). The building has 18 floors, including 4 basements. Since the ground and top floors are not public activity areas, the UTSIndoorLoc dataset covers 16 floors of the building. The total number of reference points is 1,840, covering an area of about 44,000 square meters. Among the 9494 sample data, 9107 samples are used for the training set, and the remaining 387 samples are used for the test set. A total of 589 different WiFi access points are included in the dataset.

Table 5 shows the results comparison between our best solution and methods in other literatures on the public dataset UTSIndoorLoc. The localization error is higher than [20], but the floor hit rate reaches 95.36%, which is the best result. Similar to the experimental results on the UJIIndoorLoc dataset, a balance is achieved between the floor hit rate and the localization error.

IV. CONCLUSION

When the time interval is long, the WiFi signal fluctuates strongly. This paper proposes a data enhancement method based on the nature of WiFi signals to avoid the duplication of collecting samples periodically. At the same time, the expansion of the database promotes the neural network to learn the mapping relationship between the signal fluctuation interval and the position. And SDAE is improved by adding reasonable noise to the fingerprint data so that its distribution can approximate the distribution of new fingerprint data that fluctuates due to time, and by extracting robust features from the dynamic fingerprint data to eliminate the effect of time on the WiFi signal. In this paper, experiments are carried out on three datasets, and the experimental results demonstrate the effectiveness of the data augmentation method and improved SDAE.

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